# Huawei Access Node KR-ACC-12 Optical Transport Failure Resolution SOP

**Node:** KR-ACC-12 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8258

## Description

This Standard Operating Procedure (SOP) is designed to address a Major Optical Transport Failure on Huawei Access Node KR-ACC-12 at KR Puram Access site. The failure domain is primarily caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Optical Transport Network (OTN) is a critical component of the Access Network, responsible for transmitting data over optical fibers. The OTN uses the Optical Channel Layer (OCh) to transmit data channels, which are then multiplexed and transmitted over the fiber. In the event of an SFP/XFP degradation or fiber cut, the OTN will trigger an Alarm Code ALARM_8258, indicating a Major Optical Transport Failure. This SOP outlines the diagnostic and resolution steps to isolate and fix the root cause of the failure.

## Pre-requisite Checks

- check: Dispatch remote hands to KR Puram Access site for on-site assistance, status: Completed
- check: Verify the redundant link status and ensure it is functioning properly, status: Completed

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP degradation | High |
| Fiber cuts | Medium |
| Rx/Tx power loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display optical-fiber link, Expected Output: Optical fiber link information:, Description: This command displays the link information of the optical fibers, including the SFP/XFP status, fiber status, and signal quality. This will help identify if the SFP/XFP is degraded or if there is a fiber cut.

Command: display optical-fiber alarm, Expected Output: Optical fiber alarm information:, Description: This command displays the alarm information related to the optical fibers, including the alarm code, description, and status. This will help identify if the fiber cut or SFP/XFP degradation is causing the alarm.

Command: display optical-fiber signal, Expected Output: Optical fiber signal information:, Description: This command displays the signal quality information of the optical fibers, including the Rx/Tx power, signal-to-noise ratio, and bit error rate. This will help identify if there is a problem with the signal quality.
```

## Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-fiber module {module_id}, Description: Replace the SFP/XFP module with a new one to resolve the degradation issue. Make sure to update the database with the new module ID and configuration.

Step 2: Step: Repair or replace the fiber cut, Command: repair optical-fiber {fiber_id}, Description: Repair or replace the fiber cut to restore the optical signal. Make sure to update the database with the new fiber ID and configuration.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust optical-fiber power {power_level}, Description: Adjust the Rx/Tx power to ensure optimal signal quality. Make sure to update the database with the new power level and configuration.

## Post-Check Verification

```
Command: display optical-fiber link, Expected Output: Optical fiber link information:, Description: Verify that the SFP/XFP status and fiber status are normal after the resolution steps.

Command: display optical-fiber alarm, Expected Output: Optical fiber alarm information:, Description: Verify that the alarm code and description are reset to normal after the resolution steps.

Command: display optical-fiber signal, Expected Output: Optical fiber signal information:, Description: Verify that the signal quality is optimal after the resolution steps.
```

## Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore optical-fiber module {module_id}, Description: Restore the original SFP/XFP module if the replacement module does not resolve the issue.

- Step: Restore the original fiber cut, Command: restore optical-fiber {fiber_id}, Description: Restore the original fiber cut if the repair or replacement does not resolve the issue.

- Step: Restore the original Rx/Tx power, Command: restore optical-fiber power {power_level}, Description: Restore the original Rx/Tx power if the adjustment does not resolve the issue.